

KOLBA, Russian Federation

WMO: 296750

Lat: 55.1225N

Lon: 93.3837E

Elev: 477

StdP: 95.72

Time Zone: 7.00 (E07)

Period: 90-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.8	(c) -34.3	(d) -40.1	(e) 0.1	(f) -37.6	(g) -36.7	(h) 0.1	(i) -34.1	(j) 5.3	(k) -7.9	(l) 4.2	(m) -8.9	(n) 0.0	(o) 230	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.8	(c) 27.9	(d) 18.0	(e) 25.9	(f) 17.4	(g) 24.1	(h) 16.6	(i) 19.8	(j) 25.3	(k) 18.7	(l) 23.7	(m) 17.7	(n) 22.3	(o) 1.2	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.7	22.0	16.9	12.7	20.9	15.7	11.8	20.0	59.2	25.4	55.2	23.9	51.9	22.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.3	(b) 3.3	(c) 2.5	(e) -40.5	(f) 31.9	(g) 5.5	(h) 1.6	(i) -44.5	(j) 33.0	(k) -47.7	(l) 34.0	(m) -50.9	(n) 34.9	(o) -54.9	(p) 36.1		
(a) 4.3	(b) 3.3	(c) 2.5	(e) -38.8	(f) 21.7	(g) 4.7	(h) 2.5	(i) -42.1	(j) 23.5	(k) -44.9	(l) 25.0	(m) -47.5	(n) 26.4	(o) -50.9	(p) 28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.0	-17.9	-14.2	-6.1	0.7	7.6	14.4	16.6	13.7	7.3	0.6	-8.5	-15.7	(6)
(7)		DBStd	13.18	8.86	7.38	6.46	5.10	4.68	3.63	2.80	3.36	3.90	5.33	7.47	8.42	(7)
(8)		HDD10.0	4179	864	677	498	279	104	7	0	6	99	292	554	797	(8)
(9)		HDD18.3	6731	1123	911	757	528	334	123	67	148	332	549	804	1056	(9)
(10)		CDD10.0	514	0	0	0	1	28	139	206	121	18	1	0	0	(10)
(11)		CDD18.3	24	0	0	0	0	0	6	14	4	1	0	0	0	(11)
(12)		CDH23.3	501	0	0	0	0	24	168	208	88	12	0	0	0	(12)
(13)		CDH26.7	95	0	0	0	0	2	32	38	21	2	0	0	0	(13)
(14)	Wind	WSAvg	0.7	0.5	0.7	0.9	0.9	0.7	0.5	0.4	0.5	0.7	0.6	0.6	(14)	
(15)	Precipitation	PrecAvg	562	17	15	20	40	49	72	78	89	63	49	40	27	(15)
(16)		PrecMax	746	29	32	53	80	106	124	172	177	133	99	97	49	(16)
(17)		PrecMin	419	4	5	6	9	19	15	18	42	12	17	19	5	(17)
(18)		PrecStd	78	7	7	11	19	20	29	36	34	27	20	18	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.1	3.3	12.4	19.7	27.8	30.0	30.2	29.4	24.7	17.2	8.5	1.0	(19)
(20)		MCWB	-0.9	-0.5	4.8	8.9	15.0	17.3	20.0	20.4	15.3	9.2	4.2	-1.6	(20)	
(21)		2%	DB	-1.3	0.2	8.2	15.1	24.0	27.6	27.7	25.6	19.6	13.2	5.3	-1.3	(21)
(22)		MCWB	-3.2	-2.4	2.5	6.9	13.1	17.5	18.6	18.2	13.0	7.8	2.2	-3.1	(22)	
(23)		5%	DB	-3.5	-1.4	5.4	11.8	21.1	25.5	25.7	23.2	16.3	10.4	2.8	-2.6	(23)
(24)		MCWB	-5.1	-3.6	1.1	5.5	11.9	16.6	18.2	17.2	11.4	6.8	0.7	-4.0	(24)	
(25)		10%	DB	-5.8	-3.4	2.9	8.9	18.2	23.2	24.0	20.7	13.8	8.0	1.1	-4.1	(25)
(26)		MCWB	-7.0	-5.2	-0.3	3.9	10.4	15.8	17.3	16.1	10.2	5.2	-0.5	-5.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.7	0.0	4.9	9.9	16.2	20.1	21.4	21.3	16.6	10.5	5.1	-1.4	(27)
(28)		MDCB	0.6	2.3	11.7	16.2	23.7	25.4	26.7	26.2	22.4	15.1	8.0	0.2	(28)	
(29)		2%	WB	-3.0	-2.0	2.8	7.9	14.2	18.7	20.2	19.1	13.7	8.7	2.6	-2.9	(29)
(30)		MDCB	-1.8	0.0	7.3	14.4	22.4	24.9	25.8	24.2	18.1	12.0	4.9	-1.5	(30)	
(31)		5%	WB	-5.3	-3.6	1.4	6.0	12.8	17.6	19.2	17.8	12.2	7.1	1.0	-4.0	(31)
(32)		MDCB	-3.6	-1.5	4.9	11.4	19.4	23.5	24.0	22.3	15.6	10.0	2.5	-2.6	(32)	
(33)		10%	WB	-7.0	-5.1	-0.1	4.3	11.1	16.4	18.1	16.4	10.7	5.6	-0.4	-5.2	(33)
(34)		MDCB	-5.8	-3.3	2.7	8.5	17.1	21.9	22.4	20.2	13.2	7.8	1.0	-4.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	9.4	12.1	13.5	12.1	13.0	13.2	11.8	10.9	10.5	9.2	8.5	8.8	(35)	
(36)		5% DB	MCDBR	8.7	11.7	16.7	17.7	20.1	18.8	16.0	16.2	16.5	14.2	9.1	8.5	(36)
(37)			MCWBR	7.5	9.4	11.6	10.6	11.0	9.8	8.3	9.3	9.7	8.8	6.3	7.5	(37)
(38)			MCDBR	8.8	10.8	15.3	16.0	16.4	15.2	13.9	14.6	13.8	13.0	8.1	8.9	(38)
(39)		5% WB	MCWBR	7.8	8.8	10.6	10.1	9.9	8.9	8.2	9.0	8.8	8.5	6.2	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.310	0.355	0.394	0.409	0.399	0.414	0.400	0.362	0.343	0.311	0.279	(40)	
(41)		taud	2.046	2.014	1.964	2.099	2.193	2.327	2.307	2.314	2.367	2.263	2.070	1.981	(41)	
(42)		Ebn at Noon	621	735	789	821	833	847	824	812	796	705	560	501	(42)	
(43)		Edh at Noon	67	101	136	138	133	118	119	111	92	80	66	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.90	1.91	3.23	4.25	4.95	5.91	5.22	4.29	2.86	1.55	0.80	0.56	(44)	
(45)		RadStd	0.10	0.12	0.30	0.26	0.57	0.53	0.38	0.39	0.37	0.14	0.10	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page